# Modeling Huntington's Disease in 3-D Striatal Organoids

Parish Bracha[1, 2], Kizito-Tshitoko Tshilenge[1], Melia Granath-Panelo[1], Tomoyo Sawada[3], Jennifer A. Erwin[3], Lisa M. Ellerby[1]
[1]The Buck Institute for Research on Aging, Novato, California, [2]University of California, Davis, California, [3]Lieber Institute for Brain Development, Baltimore, MD, USA

## Introduction

**Huntington's Disease (HD)**:
HD is caused by a mutation in the **huntingtin gene (HTT)** (Fig. 1).

- Abnormal expansion of CAG repeats in HTT creates polyglutamine (**polyQ**) repeats in the huntingtin protein.

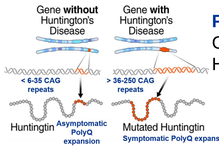



**Fig. 1** Schematic of CAG repeats in the HTT gene.

**HD:**

- Is a **fatal inherited autosomal dominant disease**.
- Has a prevalence of 5-10 individuals per 100,000 in the Caucasian population.
- Has a mean age of onset at 40 years old, with juvenile forms accounting for 5% of cases.
- Symptoms are characterized by progressive loss of motor, cognitive, and psychiatric functions.

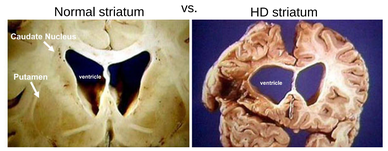



**Fig. 2** HD neurodegeneration illustrated by the atrophy of the striatum.

The **striatum**:

- Is one area of the brain dramatically affected by HD (Fig. 2).
- Is involved in motor control and reward/aversion behavior.
- Is composed of specialized inhibitory neurons called **medium spiny neurons (MSNs)**.
- Is compartmentalized into the **striosome and matrix** (Fig. 3).

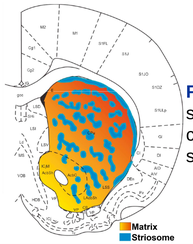



**Fig. 3** Schematic of the striosome and matrix compartments of the striatum.

Modeling HD neuropathogenesis with **induced pluripotent stem cells (iPSCs)** derived from an HD patient in two-dimensional (2-D) culture (Fig. 4).

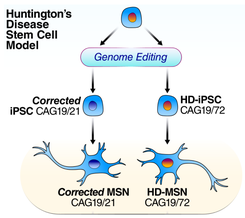
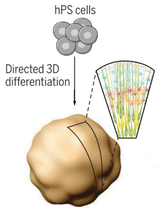



**Fig. 4** HD disease modeling using iPSCs.

The iPSC-derived three-dimensional **(3-D) brain organoid** recapitulates the complex *in vivo* organization of the human brain (Fig. 5).




In this study, we used HD and **isogenic** (C116) iPSCs to differentiate the 3-D brain organoid.

**Fig. 5** 3-D brain organoid model system.

## Hypothesis

**Background:** A previous study demonstrated that the transcription factors: **NR4A1, OLIG2, FOXF2** and **STAT1** are implicated in regulating striatal compartmentalization, more precisely, in the development of the striosome (Cirnaru et al., Manuscript under review in eLife).

**Hypothesis: The overexpression of the transcription factors NR4A1, OLIG2, FOXF2 and STAT1 in the 3-D brain organoid will generate a relevant model of striatal development.**

**Approach:** We used adenovirus-based vectors to overexpress the transcription factors in the 3-D brain organoid (Fig. 6).

**Characterization:**

- 3-D striatal organoids are characterized using **progenitor**, **pan-neuronal** and **mature** neural markers.
- **DARPP-32 and MOR** are used as markers for general striatal and striosome development, respectively.

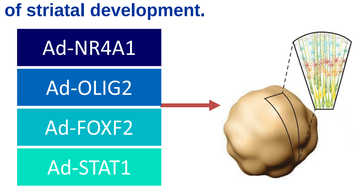
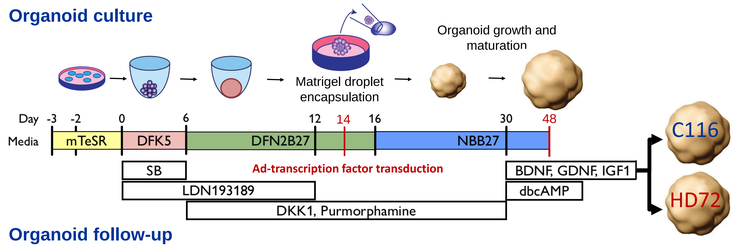


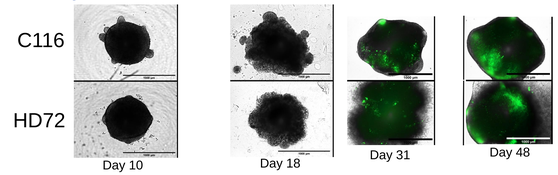

**Fig. 6** Overexpression of the transcription factors in the 3-D brain organoid.

## Methods

**Organoid culture**



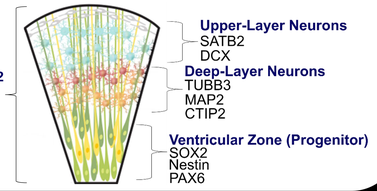

**Organoid follow-up**



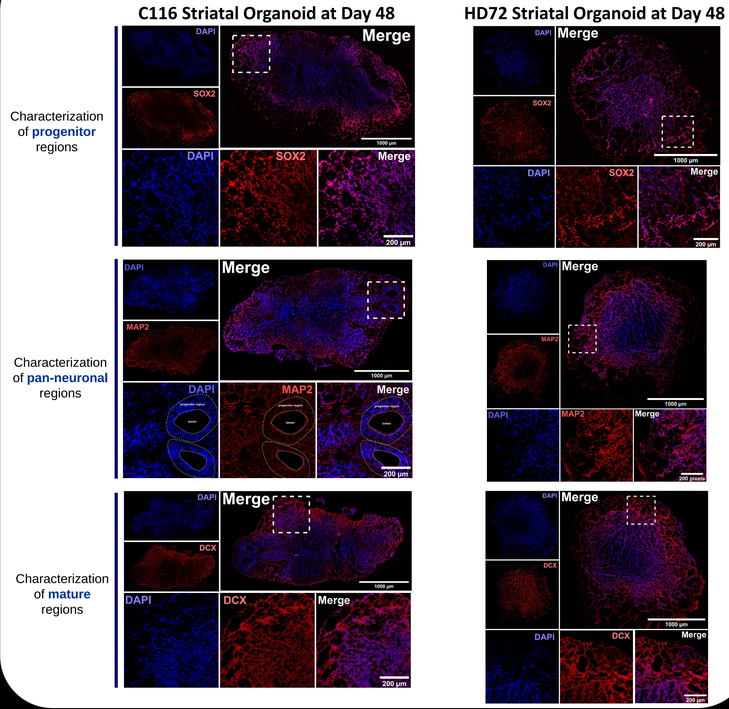

**Organoid characterization**

1. Fixation with PFA 4%
2. Immersion in sucrose 30%
3. Embedding in OCT
4. Cryosection

DARPP-32 & MOR

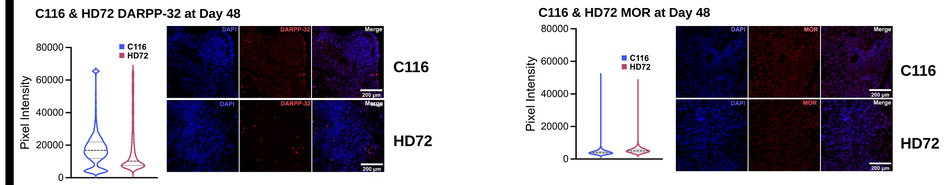


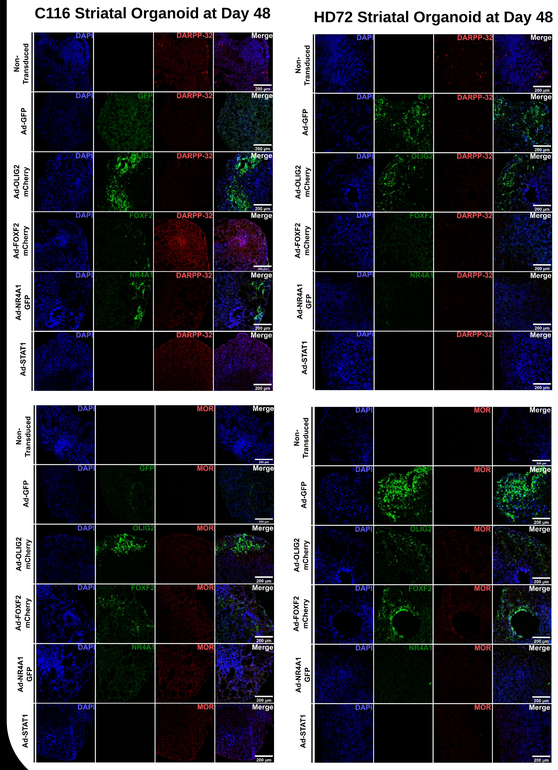

## Results

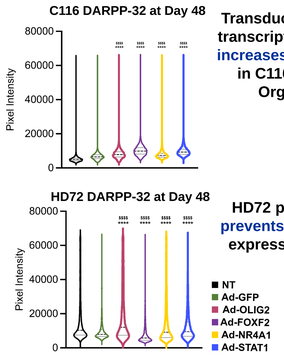



## Results

**Characterization of DARPP-32 and MOR**




**Transduction of the transcription factors increases DARPP-32 in C116 Striatal Organoid**

**HD72 phenotype prevents DARPP-32 expression levels**

**Transduction of transcription factors increases MOR in both C116 & HD72**

## Conclusions & Future Direction

**Conclusions:**

- C116 striatal organoids **recapitulate the human brain organization** in terms of progenitor, pan-neuronal, and mature regions.
- HD72 striatal organoids are **dysregulated** when compared to C116 in terms of structure, progenitor, and mature regions.
- Non-transduced HD shows **lower DARPP-32** and **increased MOR expression** compared to C116.
- The transcription factors **positively regulate MOR** levels in C116 and HD striatal organoids, which may illustrate a role in striosome development.

**Future Direction:**

- Examine NR4A1, OLIG2, FOXF2, & STAT1 in 3-6 month old striatal organoid tissue.
- Perform a loss of function of the transcription factors to evaluate their role in 3-D striatal organoids.

## Acknowledgements

Technical assistance from Kizito-Tshitoko Tshilenge and Carlos Galicia Aguirre. Supported by NIA grant R25AG048787. PB is a participant in the UC Davis Advancing Diversity in Aging Research (ADAR) Honors Program (NINDS grant R25NS112130).